Part E - Final Regression Testing / Comparison

Test Case Before	Refactoring After	Refactoring	Improvement
TC-01 Basic Insert	Functional but uses global state	Functional and encapsulated	Better modularity and safety.
TC-02 Dynamic Resize	Memory leak	Old buffer properly deleted	No resize memory leak.
TC-03 Traversal	Out-of-bounds read	Correct 0 to size−1 traversal	Eliminates garbage reads.
TC-04 Search Existing	Works	Works	Simplified to one loop.
TC-05 Search Missing	Works	Returns -1	Correct behavior maintained.
TC-06 Deletion	Works with redundant loop	Single-pass shifting	Cleaner O(n) implementation.
TC-07 Negative Index	Rejected	Rejected safely	Validation maintained.
TC-08 Upper Boundary	Possible invalid access	Rejected safely	Prevents memory corruption.
TC-09 Matrix Transpose	Works for 3×3	Works for 3×3	Cleaner implementation.
TC-10 Cleanup	Memory leak	Destructor releases memory	RAII memory safety.